Andrew Shentu

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EDUCATION

University of California, Los Angeles *Expected Dec 2027*
B.S. Electrical Engineering, GPA 3.80

- Transferred from College of San Mateo in 2025 (GPA 3.97)
- **Coursework:** Analog Electronic Circuits*, Digital Signal Processing*, Electromagnetics, Circuit Theory I & II, Signals & Systems, Digital Logic Design, Computer-Aided Design (SolidWorks) **in progress*

TECHNICAL SKILLS

Hardware	KiCad (schematic, layout, PCBA fab outputs), LTspice, SolidWorks (parts, assemblies, drawings), soldering, FDM 3D printing, laser cutting
Firmware & Software	C/C++, Python, MATLAB, QMK/VIA, Git
Lab & Manufacturing	Oscilloscope, multimeter, IEC-711 acoustic coupler, JLCPCB PCBA, overseas CNC sourcing

EXPERIENCE

Independent Keyboard Hardware Designer — *self-directed, small-batch production* *Mar 2026 – Present*

- Designed two STM32F072 keyboard PCBs in KiCad and shipped 62 boards to 26 customers across two JLCPCB assembly runs (15 in Apr 2026, 50 in Sep 2026), handling fab and assembly files, part sourcing, and shipping.
- Built the Velvet Pro board from its keyboard designer's (Ampersandy) outline; extended it into the Uni60, a universal 60% board supporting ANSI, ISO, split keys, stepped caps, and five bottom-row layouts.
- Placed 5 daisy-chained JST-SH daughterboard positions to match different cases' cable channels, plus a low-profile Molex Pico-EZmate that fits any case.
- Flashed and row/column-tested every board before shipping, with zero failures reported; open-sourced all PCB designs and QMK/VIA firmware.
- Designed a 6063 aluminum O-ring-mount case with a brass weight in SolidWorks and sourced CNC machining and anodizing overseas; fixed a ping in the 5-unit rev 2 by holding the weight with 6 M3 screws instead of 2.

PROJECTS

Tendon-Driven Bionic Hand — *UCLA engineering course project, team of 3* *Fall 2025*

- Designed the 3D-printed hand and forearm in SolidWorks, packaging 5 servos, an Arduino Uno, and a battery pack; each servo pulls a tendon that curls one finger.
- Routed the tendons along non-interfering paths through channels shaped to print without collapsing, and set the joint geometry so the thumb and index finger meet in a pinch grip.

Solar-Tracking Weather Station — *IDEA Hacks 2026, 36-hour hackathon, team of 5* *Apr 2026*

- Owned the mechanical design: a laser-cut plywood enclosure, a servo-driven turntable that turns the solar panel toward the sun, and mounting for the team's four-sensor electronics.

Multi-Driver In-Ear Monitor Crossover — *personal project* *May 2026 – Present*

- Designed a 4-way passive crossover PCB in KiCad (open-sourced as a template) that splits the audio signal across four balanced-armature drivers per side.
- Measured frequency response on an IEC-711 coupler and iterated RC values toward a target curve.

LEADERSHIP & ACTIVITIES

IEEE at UCLA, Student Project Initiative: $150 grant to build a rev 2 multi-driver IEM *Oct 2026 – Present*
Toastmasters District 4, Area Director overseeing 3 clubs; named Area Director of the Year *2023*
SCUDEM IX, Outstanding Award in a differential-equations modeling competition *2024*
First Presbyterian Church of Burlingame, Audio-Visual Assistant *2022 – 2024*